$$p_y' = p_y, \quad (2)$$

$$p_z' = p_z, \quad (3)$$

$$E'/c = \frac{E/c - (V/c)p_x}{\sqrt{1-V^2/c^2}}. \quad (4)$$

1.2 **[0,6점]**

$$p = \frac{mv}{\sqrt{1-v^2/c^2}}, \quad (5)$$

$$E = \frac{mc^2}{\sqrt{1-v^2/c^2}}. \quad (6)$$

$$inv = E^2 - p^2c^2 = m^2c^4. \quad (7)$$

제2 부. 도플러 효과와 광행차 (4,6점)

2.1 **[1,0점]** (7)

$$p = \frac{E}{c}. \quad (8)$$

$$E = \hbar\omega. \quad (9)$$

$$p_x = p\cos\varphi, \quad (10)$$

$$p_y = p\sin\varphi, \quad (11)$$

(B1.4)

$$\omega' = \omega\frac{1-V\cos\varphi/c}{\sqrt{1-V^2/c^2}}. \quad (12)$$

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2.2 **[0,4점]** (2), 8) (9)

$$\frac{\hbar\omega'}{c}\sin\varphi' = \frac{\hbar\omega}{c}\sin\varphi. \quad (13)$$

(12)

$$\sin\varphi' = \frac{\sqrt{1-V^2/c^2}\sin\varphi}{1-V\cos\varphi/c}. \quad (14)$$

(14) .

2.3 **[1,0점]** .

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,

(14)

$$\delta\varphi = \varphi' - \varphi \approx \frac{V}{c}\sin\varphi, \quad (15)$$

φ V .